

Seat Number:

National University of Singapore
Department of Mathematics
Semester 1, AY2026/2027
MA1521 Calculus for Computing
Sample Test
September 2026 — Time allowed: 2 hours

INSTRUCTIONS TO CANDIDATES

- Write down your **student number** on the right and using ink or pencil, shade the corresponding circle **completely** in the grid for each digit or letter. **DO NOT WRITE YOUR NAME!**
- Write down your **seat number** in the box on the upper right hand corner.
- This test paper consists of 24 questions, for a total of 48 marks. Including the cover page, there are 6 printed pages. Answer all 24 questions.
- This is a closed book test, but you are allowed to bring in one A4-size and double-sided helpsheet.
- The use of any form of communication or programmable computing device (e.g. mobile phone, laptop, tablet, etc.) is strictly prohibited during the test. However, you may use a non-programmable calculator.
- Write legibly. Follow instructions strictly. You may use a pen (blue or black ink) or a pencil.
- For each question, only an answer written within the given answer box will be marked; a correct and legible answer will be given 2 marks.
- This booklet will be collected at the end of the test. Do not tear off any pages of this booklet.

STUDENT NUMBER												
A												
U	☐	0	0	0	0	0	0	0	0	A	N	
A	☒	1	1	1	1	1	1	1	1	B	R	
HT	☐	2	2	2	2	2	2	2	2	E	U	
NT	☐	3	3	3	3	3	3	3	3	H	W	
		4	4	4	4	4	4	4	4	J	X	
		5	5	5	5	5	5	5	5	L	Y	
		6	6	6	6	6	6	6	6	M		
		7	7	7	7	7	7	7	7			
		8	8	8	8	8	8	8	8			
		9	9	9	9	9	9	9	9			
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Total Score:

1. Evaluate $\lim_{x \rightarrow 0} \frac{e^{2x} - \cos x - 2 \sin x}{x^2}$.

Give the answer as an integer or a fraction in lowest terms.

2. Let C be the curve defined by $x^2 - 7y^2 + y^3 = 6xy$. Find **all** points (x, y) on C where the tangent line is horizontal.

3. Let $x = x(t)$ and $y = y(t)$ be functions of t , related by the equation

$$2x^3 + 3y^2 + xy = 6.$$

Given that $x(0) = 1$, $y(0) = 1$ and $x'(0) = 4$, find $y'(0)$.

Give the answer as an integer or a fraction in lowest terms.

4. Evaluate $\int_0^{\infty} x e^{-x/2} dx$.

Give the answer as an integer or a fraction in lowest terms.

5. Consider the function f given by

$$f(x) = \int_{x^2}^{3x} \left(\frac{t}{3} + 1 \right) dt.$$

Find $f'(1)$.

Give the answer as an integer or a fraction in lowest terms.

6. Evaluate $\int_0^{\ln 3} e^x (1 + e^x)^2 dt.$

Give the answer as an integer or a fraction in lowest terms.

7. Find the value of the constant c such that the two vectors $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$ and $c\mathbf{i} + 5\mathbf{j} + \mathbf{k}$ are orthogonal.

Give the answer as an integer or a fraction in lowest terms.

8. Let $\vec{U} = 4\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$ and $\vec{V} = \mathbf{i} + 2\mathbf{j} + 2\mathbf{k}$, and suppose $\vec{Q}$ is a vector perpendicular to $\vec{V}$ such that $\vec{U} - \vec{Q}$ is parallel to $\vec{V}$. Find the vector $\vec{Q}$.

Give the answer in the form $a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$, where a, b, c are real numbers.

9. Determine the equation of the plane containing the three points $(1, 2, -1)$, $(3, 1, 0)$ and $(-1, 0, 2)$.

Give the equation in the form $ax + by + cz = 1$ where each of a, b, c is an integer or a fraction in lowest terms.

10. Let L be the line with parametric equations

$$x = t, \quad y = 2t, \quad z = 1.$$

Let Π be the plane that contains L and passes through the point $(1, 1, 3)$. Find an equation of the plane Π .

Give the answer in the form $z = ax + by + c$, where a, b, c are real constants.

11. The plane defined by the equation $2x - y + 3z = 10$ intersects the line parallel to the vector $\langle 1, 2, 1 \rangle$ and passing through the point $(1, 1, 1)$. Determine the point of the intersection.

12. Let C be the curve in the xyz -space with parametrization given by

$$\mathbf{r}(t) = (3 - t^2)\mathbf{i} + t^3\mathbf{j} + (4t - 1)\mathbf{k}.$$

Let L be the tangent line to C at the point where $t = 1$. Find the point at which L intersects the yz -plane.

Give the answer in the form $(0, b, c)$, where b, c are real numbers.

13. It is given that $f(x, y)$ is a function such that

$$\frac{\partial f}{\partial x} = 2x + y^2, \quad \frac{\partial f}{\partial y} = 2xy + 3y^2,$$

and $f(2, 1) = \frac{11}{2}$. Find the value of $f(1, 0)$.

Give the answer as an integer or a fraction in lowest terms.

14. Let $f(x, y, z) = x^2y + y^2z + z^2x$. Determine the gradient of f at the point $(1, 2, 3)$.

Give the answer in the form $\langle a, b, c \rangle$, where a, b, c are real numbers.

15. Determine the equation of the tangent plane to the surface $x^2 + y^3 + z^4 = 8$ at the point $(3, 2, 1)$.

Give the equation in the form $ax + by + cz = 1$ where each of a, b, c is an integer or a fraction in lowest terms.

16. Suppose $w = u^2 + 2uv$ where $u = xy$ and $v = -y^2$.
Determine $\frac{\partial w}{\partial x}$.

Give the answer as an expression in terms of x, y .

17. Suppose $f = f(x, y, z)$ is a function of x, y, z , and x, y, z are functions of t . At $t = 0$, it is known that x, y, z all take the value 1, while their derivatives x', y', z' with respect to t all take the value 5.

If $\nabla f(1, 1, 1) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, determine the value of $\frac{df}{dt}$ at $t = 0$.

Give the answer as an expression in terms of a, b, c .

18. Consider the function $f(x, y) = a \ln(x) + xy^2$, where a is a positive constant. It is given that the directional derivative of f at the point $(1, 2)$ in the direction of the vector $4\mathbf{i} - 3\mathbf{j}$ is equal to 2. Find the value of a .

19. Let $f(x, y) = \frac{1}{4}x^4 - \frac{1}{2}x^2 - xy + \frac{1}{2}y^2$. Find **all** critical points of $f(x, y)$.

[You do not have to classify the critical points.]

20. Consider the function $f(x, y) = e^x(x^2 - y^2)$. It is given that $(0, 0)$ is a critical point of $f(x, y)$. Determine whether this point is a local maximum point, a local minimum point or a saddle point of $f(x, y)$.

Give the answer either as **local maximum point**, **local minimum point** or **saddle point**.

21. Let $f(x, y)$ be a function such that

$$f_x = 2x + \cos(y) \quad \text{and} \quad f_y = -x \sin(y) + 2y - \pi.$$

It is given that $f(x, y)$ has a critical point at $(0, \frac{\pi}{2})$. Determine whether this point is a local maximum point, a local minimum point or a saddle point of $f(x, y)$.

Give the answer either as **local maximum point**, **local minimum point** or **saddle point**.

22. Using the method of least squares, find the cubic polynomial of the form $y = ax^3 + b$ which best fits the three data points $(-1, 1)$, $(0, -7)$ and $(1, 7)$.

Give the answer in the form $y = ax^3 + b$ with the coefficients a, b in simplest form.

23. Use Lagrange multipliers to find the maximum value of the function $f(x, y) = 2x + y$ subject to the constraint $2x^2 + y^2 = 9$.

Give the answer in the form $a\sqrt{b}$ where both a and b are positive integers, and b is as small as possible.

24. Find the point (x, y, z) which maximizes the function $f(x, y, z) = xyz$ subject to the constraints

$$x > 0, \quad y > 0, \quad z > 0 \quad \text{and} \quad x^2 + y^2 + z^2 + 2xz = 18.$$